

Option Strategies: Good Deals and Margin Calls*

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Abstract

We investigate the risk and return of a variety of trading strategies involving options on the S&P 500. Overall, we find that strategies that short options constitute very good deals. However, exploiting these good deals can be extremely difficult. Trading costs and margin requirements severely condition the implementation of the option strategies. Margin calls in particular have a double impact on trading strategies: they limit the notional amount of short-sale positions and they force investors out of trades precisely when they are losing money. These frictions limit the capacity of sophisticated investors to arbitrage away the mispricings in options markets.

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Dear Customers:

As you no doubt are aware, the New York stock market dropped precipitously on Monday, October 27, 1997. That drop followed large declines on two previous days. This precipitous decline caused substantial losses in the fund's positions, particularly the positions in puts on the Standard & Poor's 500 Index. [...] The cumulation of these adverse developments led to the situation where, at the close of business on Monday, the funds were unable to meet minimum capital requirements for the maintenance of their margin accounts. [...] We have been working with our broker-dealers since Monday evening to try to meet the funds' obligations in an orderly fashion. However, right now the indications are that the entire equity positions in the funds has been wiped out.

Sadly, it would appear that if it had been possible to delay liquidating most of the funds' accounts for one more day, a liquidation could have been avoided. Nevertheless, we cannot deal with "would have been." We took risks. We were successful for a long time. This time we did not succeed, and I regret to say that all of us have suffered some very large losses.

— Letter from Victor Niederhoffer to investors in his hedge funds

Using data on S&P 500 options from January of 1985 to May of 2004, a period that encompasses a variety of market conditions, we find that strategies involving short positions in options generally offer very high Sharpe ratios – close to 2 on an annual basis for shorting straddles and strangles. These findings are consistent with Bakshi and Kapadia (2003), Bondarenko (2003b), Coval and Shumway (2001), Driessen and Maenhout (2003a) and Jones (2004).¹ These Sharpe ratios are statistically significant even after taking into account the non-normal distribution of option returns and the possibility that the frequency of market crashes may be higher than in the sample studied. Indeed, we show that a power-utility investor would want to take a sizable position in these option strategies. Furthermore, the strategies' returns are substantially higher than warranted by common asset pricing models such as the CAPM, the Fama and French (1993), and the Leland (1999) models, and the Leland (1999) models, they cannot be explained by the hedging demands of a long-horizon investor or by reasonable expectations about the frequency of financial crashes like the one of October 1987. It seems that there are “good deals” to be had in option markets. This of course raises the question: Why are these good deals not arbitrated away?

¹Similarly, Jackwerth (2000), Rosenberg and Engle (2002) find discrepancies between the S&P 500 empirical distribution and the distribution implicit in the index option prices. Other studies, including Bakshi, Cao, and Chen (1997), Bates (2000), Benzoni (2001), Buraschi and Jackwerth (2001), Chernov and Ghysels (2000), Eraker (2004), Liu, Pan, and Wang (2005), and Pan (2002), find that option prices can be rationalized only by large volatility and jump risk premia. Evidence of misspricing is also indirectly captured by the average difference between implied and realized volatility reported by Day and Lewis (1992) and Christensen and Prabhala (1998).

Anecdotal evidence suggests that the industrial organization of the index option market is characterized by a large group of investors who buy options for insurance purposes and by a small group of market-makers who provide liquidity and take the short side of the trades (see Bates (2003)). Recent research by Garleanu, Pedersen, and Poteshman (2005) indeed shows that the net demand of private investors tends to be positive for index put options while market makers adopt the role of liquidity suppliers. In their model, wherein market makers are risk-averse and options cannot be completely hedged, prices of out-of-the-money options are set high to compensate market makers for the unhedgeable risk related to always taking the short side of a trade. This is a good argument for why market makers might not supply enough options and leave option prices too high. However, given the extraordinary Sharpe ratios offered by some option strategies, it is hard to understand why sophisticated investors (e.g., hedge funds) would not compete with market makers to provide liquidity, thereby eroding the good deals in option prices. We propose that high transaction costs and margin requirements introduce frictions in option markets that make it hard for investors to systematically sell options.

Our paper conducts a systematic analysis of the impact that margin requirements and transaction costs have on the execution of option strategies. We find that transaction costs significantly decrease the Sharpe ratios of option strategies, sometimes making them negative. We also study how margin requirements impact the returns of strategies that involve shorting options.² We find that margins affect option strategies along two dimensions: they limit the maximum amount that an investor can short, and they frequently force the investor to close down positions. Interestingly, this forced liquidations tend to happen precisely when the strategies are losing money. Margin calls also force the investor to execute a larger number of trades, increasing the importance of transaction costs. The cumulative effect

²There have been only a few studies of margin requirements in option markets. Heath and Jarrow (1987) show that the Black-Scholes model would still hold in the presence of margins. Mayhew, Sarin, and Shastri (1995) find that a decrease in the initial margin requirement is associated with an increase in the bid-ask spread and a decrease in the quote depth in the underlying market and a decrease in the option's bid-ask spread. John, Koticha, Narayanan, and Subrahmanyam (2003) show in a theoretical framework that margin requirements may increase market efficiency only when they are relatively large or small. There is a much wider literature that focuses on margins for stock trades. Day and Lewis (1997), Hardouvelis (1990), Hardouvelis and Peristiani (1992), Hardouvelis and Theodossiou (2002), Hsieh and Miller (1990), Largay III and West (1973) study the impact of changes in margin requirements on volatility. A second branch of this literature, including Figlewski and Webb (1993), Danielsen and Sorescu (2001), Jones and Lamont (2002), Lamont and Stein (2004), and Ofek, Richardson, and Whitelaw (2004), focuses explicitly on short-sale constraints and their relation to mispricings. In particular, Ofek, Richardson, and Whitelaw (2004) show how put-call parity violations are related to the cost and difficulty of short-selling the underlying.

of margin calls and transaction costs is such that, in many cases, the strategies become totally unattractive to investors. For example, the annualized Sharpe ratio of shorting a near-maturity out-of-the-money strangle decreases from 1.83 to 0.16 when trading frictions are taken into account.

Consistent with the arguments of Shleifer and Vishny (1997) and Liu and Longstaff (2004) about the limits to arbitrage, our findings explain why the good deals in options prices have not been arbitrated away and why speculative investors do not compete with market makers to provide liquidity by taking the short side of the trade in index options.

The rest of the paper is organized as follows. In Section 1 we describe the data. In Section 2 we explain the option strategies studied in the paper. In Section 3 we discuss the risk and return profile of the strategies. Section 4 analyzes the impact of transaction costs and margin requirements on the profitability. Section 5 concludes.

1 Data

We analyze two datasets of option prices. Our main tests are conducted using data provided by the *Institute for Financial Markets* for American options on S&P 500 futures traded at the Chicago Mercantile Exchange. This dataset includes daily closing prices for options and futures between January 1985 and May 2001. We also use data from *OptionMetrics* for European options on the S&P500 index which are cash settled and traded at the Chicago Board Options Exchange. This dataset includes daily closing bid and ask quotes for the period between January 1996 and May 2004.

To minimize the impact of recording errors and to guarantee homogeneity in the data we apply a series of filters. First we eliminate prices that violate arbitrage bounds. For example, we require that call prices do not fall outside the interval $(Se^{-\tau d} - Ke^{-\tau r}, Se^{-\tau d})$, where S is the value of the underlying asset, K is the option's strike price, d is the dividend yield (set to zero for futures options), r is the risk free rate, and τ is the time to expiration. Second we eliminate all observations for which the ask is lower than the bid, for which the bid is equal to zero, or for which the spread is lower than the minimum ticksize (equal to \$0.05 for options trading below \$3 and \$0.10 in any other cases). Finally we exclude all observations for which the implied Black (1976) volatility is larger than 100% or lower than 1%.

We construct monthly return time series from the closing of the first trading day of each month to the next. For options that expire within the month, the return is computed using as settlement value the opening price of the underlying on the expiration day, usually the third Friday of the month. The next section provides details on the construction of the return series.

2 Strategies

We analyze several option strategies standardized at different maturities and moneyness levels. We focus on two different maturities termed “near” (N) and “far” (F), corresponding to maturities of approximately 45 and 180 days respectively, and three different levels of moneyness, at the money (ATM), 5%, and 10% out of the money (OTM).

We consider naked and covered positions in single options, combinations of calls and puts such as straddles and strangles, and time-value strategies including calendar spreads and long-short maturity combinations. Table 1 shows the composition of each strategy.³ A naked position is formed simply by the option contract. Covered positions are portfolios composed of the option and the underlying: a covered call is formed by a long position in the underlying and a short position in a call contract. In this way, if the option ends in the money, the position can be covered by delivering the underlying. A protective put combines a long position in the underlying and a long position in a put contract. The option insures the underlying position from large negative movements. We also study strategies that involve combinations of calls and puts, such as straddles and strangles. A straddle involves buying a call and a put option with the same strike and expiration date. A strangle differs from a straddle in that the strike prices are different: buy a put with a low strike and a call with a high strike. Finally we consider time-value trades, which involve the sale and purchase of options with different maturities. In particular, we study far-near (FN) straddles and strangles. The FN straddle, for example, is formed by a long position in the far-maturity straddle and a short position in the near-maturity straddle of equal moneyness. We also

³The list of strategies that we consider in this paper is by no means exhaustive. Many more strategies are usually implemented by traders and brokers. As a reference we adopted the list of strategies that are available for trading at the London International Financial Futures Exchange (LIFFE). The list can be found at <http://www.liffe.com/products/strategies/>. For ease of exposition we report results for a subset of the strategies. Summary statistics for the strategies not reported in the paper can be obtained from the authors upon request.

analyze calendar spreads (CS), which involve selling an option and buying another at the same strike but longer maturity.

All the strategies we analyze have an initial investment of \$0. The returns shown correspond to taking a long position of \$1 and a short position of \$1 and are presented as a percentage of the \$1 notional size of the trade. If the strategy involves selling options, thus generating cash at the entering date, we invest this cash at the risk-free rate. Conversely, if the strategy requires an initial investment, we finance it by borrowing at the risk-free rate. At the end of the month, or at the option-expiration day if it comes first, we liquidate the positions and reconstruct the strategy using the contract that, at that date, has the desired approximate moneyness and maturity.

For example, consider the return of the near-maturity ATM straddle for February 1989. On the first trading day of February, the call option with moneyness closest to ATM and maturity closest to 45 days, has a price of \$5.80. The corresponding put option has a price of \$7.00. A long position in the straddle requires buying both options. In order to do so we borrow \$12.80 at the LIBOR rate, which is 0.92%. During the month, the underlying futures contract (with expiration on March 17) goes from \$298.90 to \$288.00. On the first trading day of March, the prices of the two options are \$0.30 and \$12.35 respectively. The return on the zero-cost straddle is therefore $(12.65 - 12.80)/12.80 - 0.92\% = -1.17\% - 0.92\% = -2.09\%$.

The returns of the strategies are not affected by the American nature of the options traded in the CME. We compute returns based on the prices published by the exchange. And, of course, these prices already include the early exercise premium assessed by the market participants. In fact, the results we obtain with the European options traded in the CBOE are very similar to the results obtained with the American options traded on the CME.

3 Risk and Return

In this section, we study the return and risk profile of a variety of trading strategies. Our results extend the existing literature in that we have a longer sample and study a wider set of strategies.⁴

⁴Coval and Shumway (2001) study naked calls and puts, and various straddle returns. Returns of covered positions have been studied by Merton, Scholes, and Gladstein (1978) and Merton, Scholes, and Gladstein

3.1 Summary Statistics

We start by discussing summary statistics for the monthly option returns. In Table 2, for any combination of moneyness and maturity, we tabulate the average Black and Scholes implied volatility and the average price as a percentage of the value of the underlying. This last information is essential to understand the magnitude of the portfolio weights that we will analyze in the following sections and gives us an idea of how expensive the options are. For example, the average ATM implied volatility is around 17%, while the far-maturity 5% OTM put is worth 3% of the underlying value. In general, downside protection is more expensive than upside leverage.

Table 3 reports the average, standard deviation, minimum, maximum, skewness, kurtosis, and Sharpe ratio of the each of the strategies' monthly returns. The table is divided into four panels which group strategies with similar characteristics.

The first panel of Table 3 tabulates statistics of the zero-cost naked option positions. A long position in the far-maturity ATM call rewards the investor with an average return of 13.0% per month, with a Sharpe ratio of 0.167. As expected, the strategy has positive skewness (1.383) and considerable kurtosis (5.715). Other strategies based on call contracts have higher standard deviations and lower Sharpe ratios: an example is a long position in the far-maturity 10% OTM call which has an average return of 16.4% but a standard deviation of 138% and a Sharpe ratio of 0.118. Near-maturity call options offer less attractive risk-return profiles. On the put side, average returns are negative across all maturity and moneyness combinations. Although, negative returns are to be expected given the positive returns in the S&P 500 in our sample, the magnitude of the returns is striking. Selling near-maturity 10% OTM put contracts earns 55% per month on average, with a Sharpe ratio of 0.307. However, this reward is accompanied by considerable risk: the strategy has a negative skewness of -10.716 , caused by a maximum possible loss of 22 times the notional capital of the strategy. Selling put options have lower average returns ranging from 12.5% to 48.3%, and Sharpe ratio ranging from 0.116 to 0.339.

The second panel of Table 3 tabulates statistics of covered calls and protective puts. For reference, in our sample the futures contract had a mean return of 0.8%, a standard

(1982), while Bakshi and Kapadia (2003) study delta-hedged gains. Some international evidence can be found in Gemmill and Kamiyama (2000), Driessen and Maenhout (2003a), and Driessen and Maenhout (2003b).

deviation of 4.3%, skewness of -0.804, and a Sharpe ratio of 0.189. Covered call strategies have roughly the same returns as the underlying but have lower volatility, ranging between 2.7% and 3.5%. Sharpe ratios are correspondingly higher. On the other hand, protective put strategies have lower returns, but have also positive skewness and lower kurtosis.

The third panel of Table 3 tabulates statistics of combinations of calls and puts. Straddles and strangles with short maturity offer high average returns and Sharpe ratios which are increasing with the level of moneyness: a short position in the near-maturity ATM straddle returns on average 14% per month with a Sharpe ratio of 0.273, while a short position in the near-maturity 10% OTM strangle earns an average 52.7% per month with a Sharpe ratio of 0.548.

The fourth panel of Table 3 tabulates statistics of the time-value strategies. Time-value combinations of straddles and strangles have average returns that are not as high as those previously discussed but are still considerable. For example, the far-near (FN) 10% OTM strangle has a 6.5% average return with a Sharpe ratio of 0.175, which, interestingly, is accompanied by a positive skewness of 3.095, and a minimum possible return of -58%. Calendar spreads, with the exceptions of the 10% OTM call calendar spread which on average returns 20% per month with a Sharpe ratio of 0.169, produce less impressive numbers, especially if we consider the higher moments of the return distributions.

Since the performance of the strategies is consistent in various sub-samples, for brevity of exposition, complete summary statistics are not reported in the paper. A brief discussion follows. Let us consider for example the near maturity 5% OTM put. The average return for the first half of the sample, January 1985 to March 1993, is -38.1%, while in the rest of the sample, April 1993 to May 2001, the strategy averages -52.2%. Even if we consider the more recent period that starts with the burst of the “internet bubble”, 2001 up to 2004, the return of the S&P 500 index put still averaged -30% per month in a substantially bearish market.

3.2 Statistical Significance

The investigation of the significance of the statistics reported in Table 3 is particularly difficult since the distribution of option returns is far from normal, and characterized by heavy tails and considerable skewness. For this reason, the usual asymptotic standard errors

are not suitable for inference. Instead, we base our tests on the empirical distribution of returns obtained from 1,000 non-parametric bootstrap repetitions of our sample, drawing with replacement the returns of the strategies jointly. We construct and report the 95% confidence interval or the p-value under the null hypothesis. An exact description of the bootstrap procedure followed can be found in Davison and Hinkley (1997).

In Table 4 we present 95% confidence intervals for the mean, standard deviation, and Sharpe ratio of the different strategies. We note that 21 out of 35 strategies have mean returns and Sharpe ratio statistically different from zero at the 5% level. Only two Sharpe ratios are statistically higher than the market's Sharpe ratio at the 95% confidence level: the near-maturity 5% and 10% OTM strangles. However, these are very high Sharpe ratios, especially for strategies that, as we will see, are not much correlated with the market.

3.3 The Risk-Return Tradeoff in Equilibrium

The final verdict on the matter of how much compensation options offer relative to their risks can only be given by an equilibrium asset pricing model that includes option trading. Unfortunately, it is quite difficult to derive a closed-form solution for the equilibrium expected returns of option strategies under realistic assumptions about the distribution of stock returns (an example is Jones (2004)). As an alternative we investigate how the strategies behave in the context of more standard pricing models.

As a first approximation, we compute alphas for the option strategies under the CAPM, the Fama and French (1993), and the Leland (1999) models.⁵ We report these results in Table 5. Qualitatively, the two sets of alphas (CAPM and FF) give the same indications: 21 out of 35 strategies have large and statistically significant alphas. In general, near-maturity strategies are more difficult to price, as indicated by their much lower R^2 s. These results are evidence that the returns of the option strategies are not in line with their risk according to simple asset pricing models. We also notice that the Leland's alpha is only slightly smaller than the simple CAPM alpha, with approximately the same significance

⁵Leland (1999) provides a simple correction of the CAPM which allows the computation of a robust risk measure for assets with arbitrary return distributions. This measure is based on the model proposed by Rubinstein (1976) in which a CRRA investor holds the market in equilibrium. The discount factor for this economy is the marginal utility of the investor and expected returns have a linear representation in the beta derived by Leland. Subtracting Leland's beta times the market excess return from the strategies' returns gives an estimate of the strategies' alpha that is robust to the return distribution characteristics. A similar analysis for the put returns can be found in Bondarenko (2003b).

levels. This suggests that even adjusting for the non-normal distribution of returns, the CAPM is not enough to explain options returns.

3.4 An Investment Perspective

Tables 3 and 4, together with the analysis in the previous section, suggest that large returns and Sharpe ratios, not explained by asset pricing models, can be achieved using option strategies. We now study how an investor with constant relative risk aversion (CRRA) of 2 would like to invest in these strategies.⁶ We find that the strategies produce return distributions that forbid an analysis based on the first two moments alone. High Sharpe ratios are accompanied by high skewness and kurtosis. The traditional mean-variance approach to portfolio choice cannot therefore be used to study this problem. We now turn our attention to the portfolio that maximizes the utility of the investor in the sample. This analysis can be thought of as the empirical analog of the partial equilibrium portfolio choice problem studied by Liu and Pan (2003). The analysis is similar to the study of Driessen and Maenhout (2003a) and is essentially based on the approach proposed by Brandt (1999). Note that the in-sample maximization corresponds to assuming that the investor knows the empirical return distribution. This is analogous to the more traditional assumption that the investor knows the theoretical distribution of returns to solve the portfolio problem.

We consider an investor that faces an investment opportunity set composed of three assets: the risk-free asset, the S&P 500 index (which is the underlying of the options and is also a reasonable proxy for the stock market), and each option strategy in turn. We assume that the three assets can be bought and sold without frictions. For given portfolio weights on the basis assets, we can construct a time series of returns of the resulting portfolio. We then estimate the portfolio holdings by maximizing the average utility of these monthly returns. This corresponds to solving for the optimal investment policy of a myopic investor with a horizon of one month. We use a power utility function with coefficient of risk-aversion equal to 2 and optimize it using numeric methods.

In this way, we compute a metric of the desirability of the strategies that takes into

⁶From the Merton solution for the portfolio weight invested in the market by a myopic investor, $w = \frac{\mu - r}{\gamma \sigma^2}$, and considering that in equilibrium the representative investor allocates the entire wealth to the market, $w = 1$, it is possible to obtain an estimate of γ . We choose $\gamma = 2$ which matches a market volatility of 15% and an equity premium of 4.5%.

account all the moments of the return distribution.⁷ Finally, we measure the difference (Δ CE) between the certainty equivalent of the optimal portfolio of the market, risk-free asset, and option strategy and the certainty equivalent of a reference portfolio that is uniquely composed by the market. For reference, the certainty equivalent of the market is equal to 0.9% per month in our sample period. Results are presented in Table 6.

This test serves basically two purposes. First it allows us to test how far the investor's holdings are from the equilibrium investment policy, where the whole wealth is invested in the market. We test the significance of the estimated parameters against the null hypothesis that $w_{mkt}^0 = 1$ and $w_{str}^0 = 0$. Second, the test provides us with a measure, Δ CE, of how far we are from the above mentioned equilibrium. If option returns are not commensurate with their risks, as evaluated by a CRRA utility function, then the investor would want to take advantage of them. The extent to which this happens is measured by the utility gains from the optimal investment policy.

We report weight estimates on each of the basis assets as well as the gains in certainty equivalent and Sharpe ratio for the investor. Bootstrapped p-values, under the null hypothesis that $\Delta CE^0 = 0$ and $\Delta SR^0 = 0$, are shown in brackets. In general, the investor prefers to buy far-maturity and sell near-maturity calls, and holds only short positions in put contracts. The amount invested in the market varies according to the strategy, and only in three instances it is statistically different from one. The investor prefers to optimally short-sell combinations of calls and puts, while maintaining a smaller fraction of wealth in the market. Finally, with the exception of the FN strangles and 10% OTM call calendar spread, the investor is indifferent to time-value strategies.

In terms of utility gains, some of these portfolios achieve remarkable results. On average, across all the strategies, the gain in certainty equivalent over the market is 0.6% per month. For example, by shorting 11.4% in the market and 8.4% of the near-maturity 10% OTM strangle, the investor enjoys a monthly certainty equivalent that is 2.3% higher than by holding the market. The corresponding increase in the monthly Sharpe ratio would be 0.361.

In summary, some of the option strategies are extremely appealing to our investor, who would like to allocate a significant exposure to them. This indicates that the returns

⁷This analysis is also closely related to Leland's alpha. However, that approach assumes that markets are in equilibrium (zero net investment in options) whereas we take the point of view of an investor who takes returns as given and has non zero investment in options.

of the strategies are not commensurate with their risks at least for a CRRA investor with risk aversion coefficient of 2. Of course, we cannot rule out that there may exist preferences for which the returns of options are just compensation for their level of risk. However, the magnitude of the certainty equivalent gains is such that it is likely that the search for utility functions that price all options will not be easy (see for example Driessen and Maenhout (2003a) who examine various forms of non-expected utility functions). We do not pursue that exercise in this paper but note that simply varying the coefficient of risk aversion in a power utility function will not do it.

3.4.1 Multiperiod Preferences

Since the horizon of the investment may play an important role when trading options, we also analyze the investor allocation problem in an intertemporal context. It is well known that volatility is time varying and displays mean reversion. Furthermore, volatility changes are negatively correlated with the market returns. These features are likely to induce hedging demands for an investor with a multiperiod horizon. We would expect that a long-term investor who is trying to hedge volatility risk would not want to short the hedging asset, despite its average negative return, for the benefits that such asset brings when the investment opportunity set worsens (increase in volatility).

Therefore we study how the optimal allocation to the option strategies changes when the investor is concerned with maximizing his wealth over increasing horizons. We consider the same CRRA investor as before but now assume that the investor maximizes utility with an horizon of 1 (same as before), 6, and 12 months. We report these results in Table 7. We find that increasing the horizon seems to accentuate the desire to short options. Indeed, options which are held in the short-horizon portfolios with a negative weight are more aggressively shorted at longer horizons, while options which are held with a positive weight are less sought when longer horizons are considered. We conclude that intertemporal risks are unlikely to explain option returns.

3.5 The Peso Problem in Option Returns

One serious problem in examining option returns is that a small sample may not be representative for extreme events. The “peso problem” is particularly relevant for options

because the returns on these securities, by their very nature, depend on rare events. The worst return in our sample was October of 1987, when the S&P 500 fell by 21.7%. This was clearly an outlier in our sample, 5.1 standard deviations from the monthly mean, and with the next worst month producing a return of -14.5% . We conduct a simple experiment to assess how sensitive the option returns are to having events like this happen more frequently. We study how many “October 1987” crashes in the sample would be necessary to make a CRRA investor allocate none of his wealth to the option strategies. In other words, we create artificial samples where October of 1987 happens two or three times and then we repeat the allocation exercise of section 3.4 with these modified samples.⁸ Results are presented in Table 8.

We find that the weight shorted in put options generally decreases as the number of crashes in the sample increases. However, even with three crashes in the sample, the investor still wants to short puts and the weights on the near-maturity ATM and 5% OTM puts are still negative and statistically significant. Similarly, the weight shorted of the near-maturity straddles and strangles remain significantly negative despite the increase in crash frequency. More precisely, it takes 4 additional crashes to turn positive the demand for all the option strategies. We conclude that, unless the true intensity of crashes is much higher than observed so far, the high returns of option strategies cannot be justified by a peso argument.

4 Limits to Arbitrage

Thus far, we have established that several strategies involving selling options have produced large returns even after adjusting for risk in a variety of ways. In this section, we investigate the feasibility of these strategies. In particular, we examine how trading costs and margin requirements impact the returns of the strategies.

⁸Note that this is clearly a conservative exercise since the crash of 1987 is an outlier even in the much larger history of stock market returns since the early 1800’s. See Schwert (1990) for a study of early stock market returns. October of 1987 was the fourth worst month since 1802, corresponding to the percentile 0.16 of that dataset. In our data set, that month is in the percentile 0.51. Therefore the frequency with which an event of this magnitude occurs is already three times bigger in our sample than in the last two centuries of stock market history.

4.1 Transaction Costs

Trading options can be quite expensive, not only because of the high commissions charged by brokers, but, most importantly, for the large bid-ask spreads at which options are quoted. In this section we investigate the magnitude of bid-ask spreads as well as their impact on strategy returns by analyzing the OptionMetrics database which provides the best bid and ask prices at the closing of every trading day, as well as trading volume for each option. Unfortunately, this database covers a shorter period of data than the IFM database that we have used so far in this paper. However, given that the OptionMetrics database covers the last years of the sample (1996 to 2004), the trading costs we measure are likely to be, if anything, lower than those prevailing in the first years of the IFM database.

Table 9 tabulates the mean and median bid-ask spread as a percentage of the mid-point price for the index options, and the daily mean and median volume, expressed in number of contracts, for different maturities and moneyness levels. Overall, the dollar-volume weighted average bid-ask spread is 3.9% for calls and 5.1% for puts. This is the round trip cost for an investor submitting market orders.⁹ This cost will clearly have a very substantial impact on the returns of option strategies, especially for strategies that involve high turnover. As reported by George and Longstaff (1993), the bid-ask spreads are higher for OTM options. This is not difficult to explain given that the price of an OTM contract can be as low as 5 or 10 cents. Similarly, near-maturity options tend to have higher bid-ask spreads than long-maturity contracts. For example, we find that the average round-trip cost for both near-maturity ATM calls and puts is around 5.5%. Longer maturity calls and puts with similar moneyness are quoted at a 2.6% and 3.2% spread, respectively. The spread increases steeply if we move toward OTM strikes: a 5% OTM call bears a round-trip cost equal to 11.8% when close to maturity and equal to 4.3% when far from expiration. Similarly, a 5% OTM put is quoted at 9.2% and 4.5% when close and far from maturity, respectively. Further away from the money the spreads increase even more.

Besides the large spreads, we also find that the options market can be quite thin. This raises further concerns for the potential price impact of trades. In particular, the volume for far-maturity deep OTM options is very low. This lack of liquidity is likely to impact the execution of some of the most profitable strategies. The bottom panel of Table 9 tabulates the daily mean and median number of contracts traded in different moneyness level and

⁹If the investor holds the options to maturity, only half of the cost is incurred.

maturity classes. We see that the bulk of liquidity is concentrated in the near-maturity ATM strikes. Approximately half the total volume is in very short maturity contracts (less than 45 days) and another half of this is in ATM contracts. In general there appears to be more activity in puts than calls.

In Table 10 we investigate the impact of transaction costs on the returns to short-selling put options, straddles, and strangles. We report means, standard deviations, and Sharpe ratios. In particular, we compute the strategy returns from mid price to mid price (left part of the table) as well as from bid to ask price (right part of the table) which is the relevant return for an investor selling options. We note that in the present sample the average mid-point returns are slightly higher than those reported in Table 3, due to the difference in the sample periods. Mean returns of short-selling positions in puts are 4% to 6.9% per month lower when transaction costs are considered. For example, in shorting near-maturity 10% OTM puts, the difference in average return amounts to 6.9%, which corresponds to a decrease of 0.153 in the Sharpe ratio. The trading costs impact similarly the return of straddles and strangles. For example, the bid-ask spread accounts for a loss of 4.9% in the near-maturity straddle, which corresponds to a decrease in Sharpe ratio of 0.182. The Sharpe ratio of the far-maturity straddles and strangles actually becomes negative.

Although transaction costs do not completely eliminate the abnormal profitability of some option strategies, they severely reduce it.

4.2 Margin Requirements

The most rewarding strategies all involve a short position in one or more contracts. When an investor writes an option, the broker requests a deposit in a margin account of cash or cash-equivalent instruments such as T-Bills. At the end of the day, the investor positions are marked-to-market and the net change is credited to the margin account. When the account goes below a predetermined minimum,¹⁰ the investor faces a margin call and is required to make a deposit that meets the minimum necessary. Otherwise, the option position is closed and the account is liquidated.

¹⁰The minimum that is computed when a position is opened is generally referred to as the initial margin. The initial margin is the minimum requirement for the time during which that position remains open. Every day a maintenance margin is also calculated. A margin call originates only if the maintenance margin is higher than the initial margin. After that the investor will be called upon only if the maintenance margin keeps increasing.

Options exchanges, under supervision of the SEC and the CFTC, publish in their rules book what the minimum margin requirements for customers should be. This constitutes the primary layer of the margining system. There is also a secondary layer which is constituted by the margins which are imposed on the broker/dealers themselves by the clearing house. In the US there are eleven Derivatives Clearing Organizations registered with the Commodity Futures Trading Commission. Of these, the Chicago Mercantile Exchange clears trades on futures and futures options traded at the CME, while the Options Clearing Corporation (OCC) clears trades on the stock and index options traded at the American Stock Exchange, the Boston Options Exchange, the Chicago Board Options Exchange, the International Securities Exchange, the Pacific Stock Exchange, and the Philadelphia Stock Exchange.

There are essentially three types of account which are maintained by members of a clearing house: market maker accounts, proprietary accounts, and customer accounts.¹¹ In this paper we study the customer minimum margin requirements imposed by the CBOE, and the proprietary (speculative) account margins imposed by the CME on its members. The margin requirements that are applied by the two clearing houses to members are very similar in their spirit. The CME has a system called Standard Portfolio Analysis of Risk (SPAN), while the OCC has a system called Theoretical Intermarket Margin System (TIMS). Both systems are based on scenario analysis, and in what follows we assume them to be interchangeable.¹² Therefore, the analysis of both the margins on customer accounts and on brokers' proprietary accounts should be sufficient to uncover the impact of the margining system on the key players in the option markets.

4.2.1 The CBOE Minimum Margins for Customer Accounts

The margin requirements for customers depend on the type of option strategy and on whether the short positions are covered by a matching position in the underlying asset. The margin for a naked position is determined on the basis of the option sale proceeds, plus a percentage of the value of the underlying asset, less the dollar amount by which the contract is out of

¹¹As far as margins are of concern, the difference among these accounts is that market maker accounts are margined on their net positions, meaning that short positions can be offset by long positions, while other accounts are margined on all the existing short positions.

¹²The OCC does not have any available technical documentation that could be used to reconstruct the exact functioning of the TIMS system. However conversation with OCC personnel confirmed that the system is similar to SPAN.

the money, if any.¹³ Specifically, for a naked position in a call or put option, the margin requirement at time t can be found by applying the following simple rule:

- CALL: $M_t = \max(C_t + \alpha S_t - (K - S_t | K > S_t), C_t + \beta S_t)$
- PUT: $M_t = \max(P_t + \alpha S_t - (S_t - K | S_t > K), P_t + \beta K)$

where C_t and P_t are the settlement prices announced by the exchange for that day, α and β are parameters between 0 and 1, S_t is the underlying price at the end of the day, and K is the strike price of the option.

The quantification of the parameters α and β depends on the type of underlying asset and on the investor trading in the options. These parameters are usually lower for broad based indexes. For the S&P 500, the CBOE Margin Manual specifies $\alpha = 15\%$ and $\beta = 10\%$. Nonetheless brokers may charge clients with higher margins. For example, *E-Trade* imposes margin requirements to individual investors according to the same formula but with α and β equal to 40% and 35%, respectively.

4.2.2 The CME Minimum Margins for Member Proprietary Accounts

The SPAN system is a scenario-based algorithm that computes the margins on the basis of the overall risk of a specific account. The purpose of SPAN is to find what the highest possible loss of a portfolio would be under a variety of scenarios. These scenarios are constructed by considering changes in the price of the underlying and in the level of volatility. At the end of the day, the assets in the account are re-evaluated using an option pricing model (the default model is Black (1976)) under a range of underlying price and volatility movements. The range of possible movements in the underlying security is selected by the Board of Directors and the Performance Bond Sub-Committee in order to match the 99th percentile of the historical distribution of daily price changes. The scenario losses and profits of the open positions of a particular account are then examined together and the highest possible

¹³A complete description of how to determine margin requirements for various strategies can be found in the CBOE Margin Manual, which can be downloaded from the web site: www.cboe.com/LearnCenter/pdf/margin2-00.pdf. Note also that in July 2005 the SEC approved a set of new rules regarding portfolio margining and cross-margining for index options positions of certain customers, thus making the new margining system closer to the one adopted by the CME, which will be discussed in the next section.

loss is chosen to be the minimum margin requirement for that account.¹⁴

For example, the current price ranges for the S&P 500 futures are $\pm\$80$, while the volatility range is $\pm\%5$.¹⁵ SPAN generates 14 scenarios by considering combinations of 7 price changes ($\pm\$80$, $\pm 2/3 \times \$80$, $\pm 1/3 \times \$80$, 0) and the two volatility changes. In order to account for the impact of extreme price movements on deep OTM short positions, SPAN also computes potential losses in two additional scenarios which correspond to a price change of $\pm 3 \times \$80$. In these last two scenarios, only one third of the potential loss is taken into account to determine margins.

4.2.3 Empirical Results

To explore the impact of margins on the ability of investors to execute short sales, we simulate the behavior of margin requirements in the time series of data for calls and puts of different maturities and moneyness levels. Specifically, for each trading day in the sample we calculate a “haircut” ratio, which represents the amount by which the required margin exceeds the price at which the option was written. That is, the haircut corresponds to the investor’s equity in the option position. We compute the haircut ratio as $\frac{M_t - V_0}{V_0}$, where M_t is the margin at the end of each day t , and V_0 is the value of the option sold at the beginning of the month, either C_0 or P_0 . In this way, the haircut is expressed as a percentage of the notional amount that is sold short. We compute the haircut ratio for every trading day in the month until the position is closed. The ratio changes within the month reflecting the change in the option moneyness and maturity. If the underlying asset moves against the option position, the margin requirement increases and the investor receives a margin call. For example if the investor is shorting an OTM put, a drop in the price of the underlying causes an increase in the price of the option. At the beginning of the next month a new position is opened and the coverage ratio is recalculated, obtaining in this way a continuous daily time-series for the entire length of the sample.

In what follows, we use the CBOE margins as a proxy for the margin requirement for customers, and the CME margins as a proxy for the margin requirements of a proprietary account of a clearing house member. In Table 11 we report the mean, median, standard deviation, minimum, and maximum of the haircut ratio for customers (Panel A) and

¹⁴A more detailed description of how SPAN works can be found on the CME webpage at the following URL: <http://www.cme.com/clearing/rmspan/span/compt2480.html>.

¹⁵The time series of the SCAN range parameters were obtained directly from the CME.

proprietary accounts (Panel B).

On average a customer must deposit \$7.30 as margin (in addition to the option sale proceeds) for every dollar received from shorting near-maturity ATM puts. In our sample, the maximum historical haircut ratio for those options, equals 40.9. To put this into perspective, we can interpret the inverse of the haircut ratio as the maximum percentage of the investor's wealth that could be allocated to the option trade if all the wealth was committed to the margin account. For example, to maintain an open position in the near-maturity ATM put and hence to be able to post the maximum margin call in the sample, the investor would only be able to short the option for an amount equal to 2.5% of the wealth. To short the same contract, a clearing house member would have to post on average \$2.60 per dollar of options, and \$11.50 in the worst case scenario.

Although these haircut ratios already appear quite large, bear in mind that they are calculated with the margin coefficients imposed by the exchange to large institutional investors. The margin requirements are much higher for individual investors. In those cases, it becomes almost impossible to consistently keep open a short position in OTM puts. For example, the mean haircut ratio of an investor with an E-Trade account who is shorting the same put is equal to 19.6, while the maximum value skyrockets to 60.3.

This analysis does not show that margin requirements preclude investors from selling options. It does however show that margins have a real impact for option traders. First, margins may impact the profitability of the trading strategies to the extent that they may force the investor to close out the positions when facing margin calls. Second, and more importantly, margins may limit the ability of arbitrageurs to take advantage of market inefficiencies and eventually correct option misspricings.

We study the performance of a realistic zero-cost strategy that takes into account margin calls. We assume that at the beginning of every month the investor borrows \$1 and allocates that amount to a risk-free rate account to cover the margin requirements for short selling options. We determine the notional amount of the option short sale as a percentage of the one dollar. In order to determine what percentage of the one dollar the investor would choose to short sell, we maximize the average utility of a CRRA investor with a risk aversion coefficient of 2 who can invest in the risk free rate and the option strategies.¹⁶ From

¹⁶In the previous sections, the stock index was included in the investment opportunity set. It is excluded here to make the analysis more tractable. Adding the index would add complexity without additional insight. First of all, the optimal mix between market and options changes. Because the optimal weight on the index

the optimal option weight we can compute the optimal portfolio return in the absence of margins, and hence study what impact the margining system has on the implementation of the optimal strategy. In particular, we are interested in measuring the impact of the margins on the profitability of the strategies and the extent to which they limit the exposure that an investor can take by shorting options.

In implementing the strategy with margins we assume that, during the month, the investor cannot borrow more money to face margin calls. Therefore, any margin calls are met by liquidating the investment in the risk-free asset. When these investments are not sufficient to meet the margin call, the option position itself is liquidated. After the liquidation we allow the investor to open a new position with whatever money is left. In particular, we compute the new option strategy weight in such a way that the margin due does not exceed 90% of the available wealth. The 90% level is chosen to prevent that a new margin call following a small adverse movement of the underlying leads to another immediate liquidation. At the end of the month, or at expiration day if it comes first, we subtract the interest due and compute the strategy return. In this way, returns are fully comparable to the strategy returns computed without margin requirements.

In Table 12 we report the mean return, and the Sharpe ratio of the portfolio return with and without taking into account margin calls. For each strategy we report the number of times it was necessary to recompute the initial option weight at the inception of the strategy (out), and the number of times the account was closed within the month (cl). We report the optimal weight (w_{st}) in option strategies and the average effective weight (w_{st}^*), which equals the average realized exposure once the weight rescalings made necessary by the margins are taken into account. In the last two columns we report the mean return and Sharpe ratio taking into account both margin calls and transaction costs. As a measure of transaction cost we use the average return difference reported in Table 10. We concentrate on the strategies that involve shorting puts or shorting straddles and strangles since they are the ones found to be most profitable.

can be negative, it would impose margin requirements on itself and make it impossible to distinguish the effect of the margin requirements on the options from the effect of the margin requirements on the index position. Note anyway that the weights on the option strategies do not differ much from those reported in Table 6. In general, the inclusion of the index in the portfolio would increase the impact of margins: if the market weight was positive, there would be instances in which the investor would have to partially liquidate the investment in the index to provide funding to cover margin calls. If the market position was negative, there would be additional margin requirement (Regulation T or SPAN) to maintain that short position. In either cases, OTM covered or delta-hedged positions would be affected by the margining system.

We find that the presence of margin requirements greatly impacts the profitability of the near-maturity strategies. The effect is larger when margins are considered from the perspective of a customer, but they do persist even when the lower margin requirements for a clearing house member speculative account are analyzed. In both cases the impact is larger for strategies involving deep OTM options.

For example, when we consider the customer margins (Panel A), the average return of the near ATM put goes from 2.6% per month without margin to 2.0% with margin. The corresponding Sharpe ratio decreases from 0.290 to 0.196. The impact is larger for near 10% OTM puts: the average return decreases from 1.4% to 0.5%, while the Sharpe ratio drops from 0.304 to 0.056. Similarly, the near 10% strangle average return and Sharpe ratio drop from 4.1% and 0.530 to 1.0% and 0.093 per month. Looking at the clearing house member margins (Panel B) we note that the difference in profitability is still pronounced for near-maturity options. For example, margins deduct almost 10% of the ATM put returns, while the average return for the 10% OTM put almost halves after considering margins. The same, but with a larger magnitude, is also true for the combination strategies.

The second main finding is that margins limit investors from selling the quantities of OTM options they would optimally want to sell. While the desired optimal weight for near maturity 10% OTM puts is -2.6% , the effective average strategy weight is -1.9% for a customer and -2.3% for a clearing house member. The effect is bigger when the optimal weight is larger as is the case for the 10% OTM strangle. The optimal weight is -8.0% , but the effective weight when margins are considered is -4.7% for a customer and -6.4% for a member. The difference can be explained by the number of times the strategy weight has to be rescaled at the beginning of the month (out) and by the number of times the account is closed within the trading month (cl). This second case is the factor behind the average return difference: for the near-maturity 10% strangle, the average loss in the 56 months in which the investor is forced out because of not being able to meet the margin calls is about -4% for the strategy, which means that the loss relative to the strangle price is on average about -100% .

Instead of relaying on the optimal weights obtained from numerically maximizing the investor's utility, we could also derive the maximum amount of shortable options allowed by the margin requirements. In other words, at any point in time, the amount invested is determined by how many contracts the investor is able to cover margin for: for example if the wealth is \$100 and the margin requirement for a short position equals \$20, the investor

will only be able to write five contracts. The maximum portfolio weight is then determined by dividing the dollar value of the five contracts by the wealth value. The maximum effective weight for 10% OTM puts is 5.2% for customers and 25% for clearing members. Similarly the maximum effective weights for N 10% OTM strangles are 7.1% and 32.1%, respectively. As Liu and Longstaff (2004) note, an investor would not want to invest up to the limit allowed by the margin requirements. The strategies become in fact unattractive, being characterized by small Sharpe ratios and negative certainty equivalents.

If we also take into account transaction costs, the option strategies become even less appealing to the investor. Margin calls force investors to execute more than one trade per month, increasing the impact of transaction costs. Notice from the last two columns of Table 10 that there are only few strategies left with a Sharpe ratio higher than that of the market. For example, after margin requirements and transaction costs, the Sharpe ratio of the near maturity 5% strangle drops from 0.454 to 0.080 for customers, and to 0.210 for clearing members.

So far however, in maximizing the utility, we have not taken into account the fact that margins have to be maintained if the optimal portfolio weight is indeed negative. Up to now, margins were considered only *ex post*. Therefore to study how margins affect the investor's trading behavior we repeat the numerical optimization at the base of the results of Table 12 and compute the portfolio weights when the strategies' returns are constrained by the margin calls.¹⁷ In Table 13 we report the optimal option strategy weight and the certainty equivalent of the optimal portfolio in the three possible scenarios: no margin, CBOE margins, and CME margins. We find that, after taking into account the margin requirements, our power utility investor substantially reduces the proportion of wealth invested in the strategies. This is the case under both margining systems. The amount by which the portfolio weights changes is however greater in the case of the CBOE margins. Moreover, independently of which system is considered, the change in the optimal investment policy is larger for OTM strategies than for ATM strategies. The margin requirement constraint is clearly binding. Note that the impact on the investor utility, expressed by the certainty equivalent, is sometimes quite dramatic. For deep out of the money options the utility loss can amount to almost 50%.

Overall, the evidence presented in this section suggests that the margin requirements

¹⁷As before the investment opportunity set is composed only by the risk-free rate and by the option strategy. As mentioned before the market is not included because of the potential noise it would add to the analysis of the impact of margin calls.

adopted by the exchanges impede the ability of investors to short sell options, especially near-maturity contracts. The existence of frictions on the sell-side of the market has the effect of making options more expensive. Therefore, barriers to selling options allow misspricings to persist, diminishing the potential benefits that option markets might bring on risk sharing among investors.

5 Conclusion

This paper documents the performance of a variety of option strategies on the S&P 500 from 1985 to 2004. Consistent with previous research, we find that a number of strategies that involve shorting options have offered extremely high returns. These returns are hard to justify as compensation for risk, even after taking into account the nonlinear nature of option risks and their exposure to infrequent jump risks.

Transaction costs can be very substantial in option markets, especially for short-maturity out-of-the-money options which are thinly traded. These transaction costs severely reduce the profitability of option strategies and, in some cases, turn their returns negative. Most importantly, the margin requirements imposed by option exchanges can have a substantial impact on the strategies. We find that margin calls frequently force investors to close out short option positions and that this happens when those positions are suffering large losses.

Our results support the existence of large misspricings in option markets which cannot be arbitrated away due to a combination of high transaction costs and heavy margin requirements. These misspricings may seriously blunt the effectiveness of option markets for risk sharing among investors. The large disparities between the profitability of option strategies before and after taking margin requirements into account raise concerns about the formulas used by the exchanges to calculate margin requirements. Although margins may be effective in limiting counterparty default risk, they also impose a friction that may help perpetuate large misspricings.

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Table 1: Construction of Option Strategies

This table details the composition of the option strategies. We focus on two different maturities, near (N) and far (F), corresponding to approximately 45 and 180 days to maturity, and three levels of moneyness corresponding to at the money (ATM), and out of the money by 5% and 10%. All strategies have zero initial investment. If the option position generates cash at the entering date, that cash is reinvested at the risk-free rate. Conversely, if the strategy requires an initial cash outlay, it is financed by borrowing at the risk-free rate. At the end of the month, or at the option liquidation day if it comes first, the position is liquidated and rolled over into the contracts that have the desired moneyness and maturity at the time. Prices are described as a function of moneyness and maturities: $price = f(mon, mat)$, with moneyness for in-the-money options denoted with a negative sign. For example, the price of the far-maturity, 5% OTM call is $c(5, F)$ and the price of a near-maturity. The level of the S&P 500 index is denoted s .

	S&P	near		far		risk-free
		call	put	call	put	
Cov Call F ATM	s			$-c(0, F)$		$-s+c(0, F)$
Cov Call N ATM	s	$-c(0, N)$				$-s+c(0, N)$
Cov Call F 5%	s			$-c(5, F)$		$-s+c(5, F)$
Cov Call N 5%	s	$-c(5, N)$				$-s+c(5, N)$
Prot Put F ATM	s		$p(0, N)$			$-s-p(0, N)$
Prot Put N ATM	s				$p(0, F)$	$-s-p(0, F)$
Prot Put F 5%	s		$p(5, N)$			$-s-p(5, N)$
Prot Put N 5%	s				$p(5, F)$	$-s-p(5, F)$
Straddle F ATM				$c(0, F)$	$p(0, F)$	$-c(0, F)-p(0, F)$
Straddle N ATM		$c(0, N)$	$p(0, N)$			$-c(0, N)-p(0, N)$
Strangle F 5%				$c(5, F)$	$p(5, F)$	$-c(5, F)-p(5, F)$
Strangle N 5%		$c(5, N)$	$p(5, N)$			$-c(5, N)-p(5, N)$
Strangle F 10%				$c(10, F)$	$p(10, F)$	$-c(10, F)-p(10, F)$
Strangle N 10%		$c(10, N)$	$p(10, N)$			$-c(10, N)-p(10, N)$
LS straddle ATM		$-c(0, N)$	$-p(0, N)$	$c(0, F)$	$p(0, F)$	$c(0, N)+p(0, N)-c(0, F)-p(0, F)$
LS strangle 5%		$-c(5, N)$	$-p(5, N)$	$c(5, F)$	$p(5, F)$	$c(5, N)+p(5, N)-c(5, F)-p(5, F)$
LS strangle 10%		$-c(10, N)$	$-p(10, N)$	$c(10, F)$	$p(10, F)$	$c(10, N)+p(10, N)-c(10, F)-p(10, F)$
CS Call ATM		$-c(0, N)$		$c(0, F)$		$c(0, N)-c(0, F)$
CS Put ATM			$-p(0, N)$		$p(0, F)$	$p(0, N)-p(0, F)$
CS Call 5%		$-c(5, N)$		$c(5, F)$		$c(5, N)-c(5, F)$
CS Put 5%			$-p(5, N)$		$p(5, F)$	$p(5, N)-p(5, F)$
CS Call 10%		$-c(10, N)$		$c(10, F)$		$c(10, N)-c(10, F)$
CS Put 10%			$-p(10, N)$		$p(10, F)$	$p(10, N)-p(10, F)$

Table 2: Descriptive Statistics of Option Data

In this table we report the average Black and Scholes implied volatility as well as the average ratio of the option price to the value of the underlying index for calls and puts on the S&P500 futures. We focus on two different maturities, near (N) and far (F), corresponding to approximately 45 and 180 days to maturity, respectively. We report statistics for option at the money (ATM), and out of the money by 5% and 10%. Options and futures closing prices were sampled monthly between January 1985 and May 2001. The data is provided by the Chicago Mercantile Exchange through the Institute for Financial Markets. All options are American.

	PUT			CALL		
	10%	5%	ATM	ATM	5%	10%
	<i>imp vol</i>			<i>imp vol</i>		
F	0.197	0.184	0.175	0.155	0.150	0.146
N	0.239	0.203	0.172	0.164	0.154	0.153
	<i>ratio</i>			<i>ratio</i>		
F	0.019	0.028	0.038	0.030	0.021	0.016
N	0.005	0.010	0.023	0.020	0.006	0.003

Table 3: Returns of Option Strategies

This table reports summary statistics of returns of the various option strategies described in Table 1: average, standard deviation, minimum, maximum, skewness, kurtosis, and Sharpe ratio. Options and futures closing prices were sampled monthly between January 1985 and May 2001. The data is provided by the Chicago Mercantile Exchange through the Institute for Financial Markets. All options are American. For comparison, the near-maturity S&P 500 futures contract has a mean return of 1.2%, a standard deviation of 4.3%, skewness of -0.804, and a Sharpe ratio of 0.189.

	mean	std	min	max	skew	kurt	SR
Call F ATM	0.130	0.775	-0.984	3.636	1.383	5.715	0.167
Call N ATM	0.249	1.810	-1.007	17.175	5.065	42.470	0.138
Call F 5%	0.097	0.840	-0.984	3.636	1.286	4.907	0.115
Call N 5%	-0.061	1.911	-1.007	17.175	4.958	38.130	-0.032
Call F 10%	0.164	1.384	-0.995	13.995	5.665	52.867	0.118
Call N 10%	-0.131	2.191	-1.007	16.743	4.824	30.226	-0.060
Put F ATM	-0.125	0.605	-0.966	3.798	2.871	15.775	-0.207
Put N ATM	-0.292	0.991	-1.007	5.531	2.494	10.945	-0.294
Put F 5%	-0.150	0.688	-0.966	5.733	4.226	31.509	-0.218
Put N 5%	-0.483	1.426	-1.007	12.162	6.373	50.214	-0.339
Put F 10%	-0.133	1.142	-0.966	13.359	8.852	101.409	-0.116
Put N 10%	-0.550	1.793	-1.007	22.165	10.716	132.973	-0.307
Cov Call F ATM	0.008	0.031	-0.191	0.125	-1.550	12.355	0.243
Cov Call N ATM	0.007	0.027	-0.184	0.077	-2.304	15.950	0.265
Cov Call F 5%	0.008	0.034	-0.201	0.125	-1.436	10.729	0.227
Cov Call N 5%	0.009	0.035	-0.207	0.077	-1.700	9.946	0.243
Prot Put F ATM	0.004	0.027	-0.065	0.098	0.352	4.015	0.147
Prot Put N ATM	0.003	0.023	-0.072	0.102	0.904	5.173	0.131
Prot Put F 5%	0.005	0.031	-0.100	0.102	0.030	3.740	0.145
Prot Put N 5%	0.005	0.032	-0.097	0.114	0.108	3.448	0.154
Straddle F ATM	-0.056	0.273	-0.541	1.605	2.722	15.523	-0.205
Straddle N ATM	-0.140	0.515	-1.008	2.099	1.320	6.061	-0.273
Strangle F 5%	-0.058	0.326	-0.645	2.180	2.454	14.594	-0.178
Strangle N 5%	-0.398	0.850	-1.008	5.072	3.480	19.188	-0.468
Strangle F 10%	-0.047	0.470	-0.761	3.157	2.990	17.666	-0.101
Strangle N 10%	-0.527	0.961	-1.008	8.328	5.861	47.284	-0.548
FN straddle ATM	0.023	0.239	-0.938	1.057	-0.522	6.329	0.097
FN strangle 5%	0.035	0.232	-0.875	1.264	0.887	8.278	0.151
FN strangle 10%	0.065	0.373	-0.589	2.701	3.095	19.597	0.175
CS Call ATM	-0.017	0.766	-4.928	3.311	-1.752	17.198	-0.022
CS Put ATM	0.023	0.425	-2.226	1.619	-1.346	11.957	0.053
CS Call 5%	0.076	0.601	-3.623	2.741	-0.400	11.796	0.126
CS Put 5%	-0.009	0.321	-1.288	1.784	1.127	8.845	-0.030
CS Call 10%	0.204	1.208	-2.167	14.146	8.033	91.674	0.169
CS Put 10%	0.007	0.566	-1.065	4.574	4.099	28.996	0.013

Table 4: Bootstrapped Confidence Intervals

This table reports 95% bootstrap confidence intervals three of the summary statistics of returns of the various option strategies reported in Table 3: the average return, the standard deviation of returns, and the Sharpe ratio. Options and futures closing prices were sampled monthly between January 1985 and May 2001. The data is provided by the Chicago Mercantile Exchange through the Institute for Financial Markets. All options are American.

	mean [95%]		std [95%]		<i>SR</i> [95%]	
Call F ATM	0.027	0.238	0.658	0.888	0.040	0.287
Call N ATM	0.034	0.506	1.144	2.620	0.027	0.230
Call F 5%	-0.019	0.219	0.727	0.946	-0.026	0.240
Call N 5%	-0.298	0.197	1.210	2.705	-0.235	0.083
Call F 10%	-0.002	0.362	0.828	2.005	-0.002	0.209
Call N 10%	-0.390	0.167	1.367	2.969	-0.279	0.060
Put F ATM	-0.203	-0.029	0.441	0.769	-0.439	-0.039
Put N ATM	-0.426	-0.151	0.770	1.209	-0.537	-0.126
Put F 5%	-0.241	-0.039	0.451	0.951	-0.508	-0.042
Put N 5%	-0.661	-0.273	0.627	2.040	-1.062	-0.137
Put F 10%	-0.267	0.063	0.470	1.832	-0.545	0.033
Put N 10%	-0.742	-0.252	0.524	2.987	-1.416	-0.081
Cov Call F ATM	0.003	0.012	0.024	0.038	0.075	0.454
Cov Call N ATM	0.003	0.011	0.021	0.035	0.088	0.508
Cov Call F 5%	0.002	0.012	0.026	0.041	0.063	0.424
Cov Call N 5%	0.003	0.013	0.028	0.043	0.072	0.446
Prot Put F ATM	0.000	0.008	0.023	0.029	0.001	0.278
Prot Put N ATM	-0.000	0.006	0.020	0.026	-0.017	0.262
Prot Put F 5%	-0.000	0.009	0.026	0.033	-0.002	0.288
Prot Put N 5%	0.000	0.010	0.028	0.036	0.004	0.299
Straddle F ATM	-0.091	-0.014	0.203	0.346	-0.430	-0.045
Straddle N ATM	-0.211	-0.069	0.430	0.589	-0.469	-0.125
Strangle F 5%	-0.102	-0.009	0.247	0.415	-0.387	-0.022
Strangle N 5%	-0.512	-0.274	0.592	1.079	-0.833	-0.263
Strangle F 10%	-0.110	0.023	0.333	0.600	-0.310	0.040
Strangle N 10%	-0.644	-0.382	0.495	1.393	-1.284	-0.274
FN straddle ATM	-0.010	0.057	0.199	0.275	-0.040	0.252
FN strangle 5%	0.003	0.071	0.188	0.276	0.017	0.293
FN strangle 10%	0.014	0.120	0.265	0.483	0.046	0.273
CS Call ATM	-0.125	0.083	0.552	0.966	-0.150	0.121
CS Put ATM	-0.041	0.085	0.322	0.514	-0.084	0.223
CS Call 5%	-0.009	0.162	0.474	0.736	-0.014	0.292
CS Put 5%	-0.053	0.035	0.263	0.385	-0.180	0.101
CS Call 10%	0.062	0.391	0.596	1.873	0.086	0.289
CS Put 10%	-0.067	0.093	0.365	0.761	-0.167	0.128

Table 5: Asset Pricing Model Tests

This table reports CAPM and Fama-French three-factor model alphas and adjusted R-squared coefficients as well as tests of Rubinstein (1976) nonlinear CAPM model for the option strategies in Table 3. As a test for the Rubinstein's model we measure the risk of each strategy using Leland (1999)'s robust beta and report the corresponding alpha. In computing the alpha we use a risk-aversion coefficient of 2. Bootstrapped p-values are shown in brackets. Options and futures closing prices were sampled monthly between January 1985 and May 2001. The data is provided by the Chicago Mercantile Exchange through the Institute for Financial Markets. All options are American.

	α_{capm}		R^2	α_{ff}		R^2	α_{tel}		R^2
Call F ATM	0.034	[0.44]	0.542	0.027	[0.57]	0.620	0.042	[0.35]	0.467
Call N ATM	0.074	[0.51]	0.280	0.065	[0.65]	0.320	0.089	[0.41]	0.233
Call F 5%	-0.001	[0.93]	0.522	-0.011	[0.70]	0.601	0.008	[0.95]	0.447
Call N 5%	-0.220	[0.06]	0.218	-0.231	[0.04]	0.262	-0.204	[0.07]	0.176
Call F 10%	0.060	[0.58]	0.225	0.010	[0.90]	0.371	0.071	[0.50]	0.191
Call N 10%	-0.317	[0.03]	0.175	-0.326	[0.02]	0.197	-0.298	[0.03]	0.138
Put F ATM	-0.070	[0.00]	0.740	-0.061	[0.02]	0.747	-0.066	[0.01]	0.821
Put N ATM	-0.181	[0.00]	0.615	-0.175	[0.00]	0.617	-0.174	[0.00]	0.676
Put F 5%	-0.078	[0.02]	0.684	-0.063	[0.09]	0.699	-0.070	[0.06]	0.801
Put N 5%	-0.353	[0.00]	0.366	-0.331	[0.01]	0.374	-0.335	[0.01]	0.461
Put F 10%	-0.039	[0.56]	0.467	-0.004	[0.83]	0.507	-0.018	[0.74]	0.620
Put N 10%	-0.424	[0.01]	0.310	-0.378	[0.03]	0.362	-0.387	[0.03]	0.460
Cov Call F ATM	0.003	[0.02]	0.746	0.003	[0.03]	0.753	0.002	[0.03]	0.793
Cov Call N ATM	0.005	[0.00]	0.193	0.005	[0.00]	0.194	0.004	[0.00]	0.234
Cov Call F 5%	0.002	[0.05]	0.780	0.002	[0.10]	0.792	0.002	[0.09]	0.819
Cov Call N 5%	0.005	[0.00]	0.167	0.005	[0.00]	0.176	0.005	[0.00]	0.197
Prot Put F ATM	0.000	[0.97]	0.609	0.000	[0.86]	0.681	0.000	[0.84]	0.544
Prot Put N ATM	0.001	[0.08]	0.132	0.001	[0.09]	0.164	0.001	[0.08]	0.112
Prot Put F 5%	-0.000	[0.79]	0.717	-0.000	[0.74]	0.777	-0.000	[0.91]	0.662
Prot Put N 5%	0.002	[0.04]	0.136	0.002	[0.05]	0.164	0.002	[0.04]	0.123
Straddle F ATM	-0.041	[0.06]	0.090	-0.039	[0.05]	0.148	-0.036	[0.09]	0.159
Straddle N ATM	-0.121	[0.00]	0.042	-0.123	[0.00]	0.098	-0.114	[0.00]	0.077
Strangle F 5%	-0.047	[0.08]	0.035	-0.044	[0.11]	0.101	-0.041	[0.13]	0.084
Strangle N 5%	-0.365	[0.00]	0.045	-0.361	[0.00]	0.074	-0.349	[0.00]	0.099
Strangle F 10%	-0.037	[0.31]	0.014	-0.027	[0.44]	0.086	-0.027	[0.43]	0.054
Strangle N 10%	-0.495	[0.00]	0.033	-0.486	[0.00]	0.081	-0.473	[0.00]	0.096
FN straddle ATM	0.020	[0.28]	0.006	0.023	[0.18]	0.012	0.019	[0.30]	0.007
FN strangle 5%	0.030	[0.08]	0.017	0.028	[0.11]	0.022	0.028	[0.09]	0.026
FN strangle 10%	0.058	[0.01]	0.011	0.062	[0.01]	0.026	0.059	[0.01]	0.010
CS Call ATM	-0.004	[0.95]	0.008	-0.011	[0.83]	0.012	-0.004	[0.95]	0.008
CS Put ATM	0.008	[0.81]	0.035	0.017	[0.51]	0.046	0.007	[0.84]	0.039
CS Call 5%	0.033	[0.36]	0.150	0.025	[0.51]	0.185	0.035	[0.35]	0.138
CS Put 5%	0.011	[0.61]	0.120	0.010	[0.63]	0.139	0.007	[0.69]	0.084
CS Call 10%	0.125	[0.04]	0.130	0.089	[0.13]	0.273	0.130	[0.04]	0.113
CS Put 10%	0.042	[0.23]	0.116	0.045	[0.23]	0.117	0.040	[0.27]	0.100

Table 6: Optimal Investment in Option Strategies

This table summarizes the optimal portfolio of the market, the risk-free asset, and each of the option strategies in Table 3. The portfolio is found by numerically maximizing the average in-sample utility of a power-utility investor with CRRA equal to 2. The table reports the weights on the market and on each of the option strategies, the optimal number of option contracts per unit of the S&P 500 index, the increase in certainty equivalent for the investor relative to investing 100% of the wealth in the market alone, and the corresponding increase in Sharpe ratio. Bootstrapped p-values, under the null hypothesis that $w_{mkt}^0 = 1$, $w_{str}^0 = 0$, $\Delta CE^0 = 0$, and $\Delta SR^0 = 0$, are shown in brackets. Options and futures closing prices were sampled monthly between January 1985 and May 2001. The data is provided by the Chicago Mercantile Exchange through the Institute for Financial Markets. All options are American. The market is proxied by the CRSP value-weighted portfolio. For comparison, the market's certainty equivalent and Sharpe ratio are equal to 0.7% and 0.189 per month, respectively.

	w_{mk}		w_{str}		# str	ΔCE $\times 100$		ΔSR	
Call F ATM	0.754	[0.87]	0.100	[0.25]	3.4	0.369	[0.18]	0.008	[0.34]
Call N ATM	1.185	[0.75]	0.038	[0.30]	1.9	0.313	[0.23]	-0.003	[0.42]
Call F 5%	1.459	[0.62]	0.023	[0.79]	1.1	0.153	[0.28]	-0.003	[0.41]
Call N 5%	2.068	[0.12]	-0.027	[0.11]	-4.2	0.437	[0.16]	0.043	[0.22]
Call F 10%	1.249	[0.71]	0.040	[0.43]	2.5	0.275	[0.23]	-0.005	[0.46]
Call N 10%	2.122	[0.10]	-0.030	[0.04]	-9.4	0.626	[0.12]	0.062	[0.15]
Put F ATM	-1.755	[0.07]	-0.247	[0.01]	-6.5	0.927	[0.05]	0.119	[0.07]
Put N ATM	-1.715	[0.06]	-0.167	[0.00]	-7.2	1.688	[0.01]	0.172	[0.02]
Put F 5%	-0.236	[0.30]	-0.105	[0.11]	-3.8	0.478	[0.14]	0.092	[0.15]
Put N 5%	0.181	[0.41]	-0.046	[0.01]	-4.6	1.012	[0.06]	0.179	[0.06]
Put F 10%	1.714	[1.00]	0.001	[0.87]	0.1	0.135	[0.27]	-0.001	[0.34]
Put N 10%	0.310	[0.32]	-0.026	[0.06]	-5.2	0.616	[0.11]	0.162	[0.09]
Straddle F ATM	0.936	[0.96]	-0.168	[0.15]	1.5	0.415	[0.17]	0.059	[0.20]
Straddle N ATM	0.691	[0.85]	-0.174	[0.01]	0.9	1.137	[0.05]	0.117	[0.07]
Strangle F 5%	1.104	[0.74]	-0.117	[0.21]	0.5	0.349	[0.19]	0.051	[0.23]
Strangle N 5%	0.060	[0.40]	-0.118	[0.00]	-1.7	2.473	[0.00]	0.289	[0.00]
Strangle F 10%	1.495	[0.44]	-0.029	[0.61]	1.2	0.162	[0.28]	0.012	[0.34]
Strangle N 10%	-0.114	[0.58]	-0.084	[0.00]	-3.7	2.431	[0.00]	0.361	[0.00]
FN straddle ATM	1.579	[0.38]	0.157	[0.31]	6.5	0.287	[0.23]	0.018	[0.33]
FN strangle 5%	1.342	[0.57]	0.258	[0.11]	8.0	0.491	[0.15]	0.039	[0.24]
FN strangle 10%	1.381	[0.51]	0.294	[0.01]	11.1	0.936	[0.07]	0.049	[0.17]
CS Call ATM	1.674	[0.31]	-0.005	[0.95]	-0.5	0.136	[0.30]	0.000	[0.42]
CS Put ATM	1.646	[0.34]	0.017	[0.86]	1.2	0.141	[0.30]	0.001	[0.39]
CS Call 5%	1.431	[0.60]	0.066	[0.34]	5.1	0.262	[0.24]	0.008	[0.34]
CS Put 5%	1.697	[0.28]	0.019	[0.83]	1.3	0.139	[0.31]	0.002	[0.38]
CS Call 10%	0.820	[0.84]	0.148	[0.03]	8.2	0.958	[0.08]	0.007	[0.31]
CS Put 10%	1.795	[0.22]	0.062	[0.33]	4.5	0.248	[0.23]	0.016	[0.35]

Table 7: Intertemporal Asset Pricing Model Tests

This table reports results obtained repeating the CRRA utility maximization of Table 6 ($\gamma = 2$) and allowing the horizon of the investor to be longer than the monthly rebalancing frequency of the portfolio. We consider horizons of 6 and 12 months and reproduce the results for the 1-month horizon for comparison. The table reports the weight on each of the option strategies, and the increase in certainty equivalent for the investor relative to investing 100% of the wealth in the market alone. Bootstrapped p-values, under the null hypothesis that $w_{mkt}^0 = 1$, $w_{str}^0 = 0$, and $\Delta CE^0 = 0$, are shown in brackets. Options and futures closing prices were sampled monthly between January 1985 and May 2001. The data is provided by the Chicago Mercantile Exchange through the Institute for Financial Markets. All options are American. For comparison, the market certainty equivalent is equal to 0.7%, 4.5%, and 9.1% at the 1-, 6-, and 12-month horizons, respectively.

	1 month						6 month						12 month					
	w_{mkt}		w_{str}		ΔCE $\times 100$		w_{mkt}		w_{str}		ΔCE $\times 100$		w_{mkt}		w_{str}		ΔCE $\times 100$	
Call F ATM	0.754	[0.87]	0.100	[0.25]	0.369	[0.18]	1.025	[0.88]	0.082	[0.28]	0.353	[0.17]	1.391	[0.88]	0.064	[0.29]	0.397	[0.18]
Call N ATM	1.185	[0.75]	0.038	[0.30]	0.313	[0.23]	1.307	[0.79]	0.040	[0.31]	0.356	[0.22]	1.521	[0.75]	0.041	[0.32]	0.480	[0.22]
Call F 5%	1.459	[0.62]	0.023	[0.79]	0.153	[0.28]	1.645	[0.61]	0.014	[0.78]	0.181	[0.28]	2.064	[0.58]	-0.005	[0.78]	0.282	[0.28]
Call N 5%	2.068	[0.12]	-0.027	[0.11]	0.437	[0.16]	2.149	[0.11]	-0.030	[0.11]	0.553	[0.15]	2.408	[0.11]	-0.030	[0.11]	0.677	[0.15]
Call F 10%	1.249	[0.71]	0.040	[0.43]	0.275	[0.23]	1.356	[0.72]	0.042	[0.43]	0.336	[0.22]	1.604	[0.70]	0.035	[0.43]	0.399	[0.22]
Call N 10%	2.122	[0.10]	-0.030	[0.04]	0.626	[0.12]	2.256	[0.10]	-0.033	[0.04]	0.783	[0.10]	2.510	[0.10]	-0.031	[0.05]	0.889	[0.11]
Put F ATM	-1.755	[0.07]	-0.247	[0.01]	0.927	[0.05]	-2.793	[0.10]	-0.321	[0.02]	1.144	[0.05]	-2.340	[0.12]	-0.306	[0.02]	1.283	[0.04]
Put N ATM	-1.715	[0.06]	-0.167	[0.00]	1.688	[0.01]	-2.679	[0.07]	-0.211	[0.00]	2.334	[0.01]	-2.729	[0.08]	-0.235	[0.00]	2.929	[0.01]
Put F 5%	-0.236	[0.30]	-0.105	[0.11]	0.478	[0.14]	-0.725	[0.32]	-0.131	[0.12]	0.562	[0.14]	-0.317	[0.33]	-0.120	[0.12]	0.656	[0.14]
Put N 5%	0.181	[0.41]	-0.046	[0.01]	1.012	[0.06]	0.257	[0.42]	-0.044	[0.01]	1.019	[0.05]	0.531	[0.44]	-0.044	[0.02]	1.204	[0.06]
Put F 10%	1.714	[1.00]	0.001	[0.87]	0.135	[0.27]	2.108	[0.98]	0.011	[0.88]	0.190	[0.25]	2.307	[0.99]	0.009	[0.86]	0.296	[0.25]
Put N 10%	0.310	[0.32]	-0.026	[0.06]	0.616	[0.11]	0.025	[0.33]	-0.029	[0.08]	0.649	[0.10]	0.501	[0.35]	-0.025	[0.09]	0.722	[0.10]
Straddle F ATM	0.936	[0.96]	-0.168	[0.15]	0.415	[0.17]	1.025	[0.93]	-0.167	[0.15]	0.414	[0.17]	1.304	[0.92]	-0.159	[0.18]	0.537	[0.17]
Straddle N ATM	0.691	[0.85]	-0.174	[0.01]	1.137	[0.05]	0.618	[0.86]	-0.189	[0.01]	1.312	[0.05]	1.012	[0.88]	-0.190	[0.01]	1.602	[0.05]
Strangle F 5%	1.104	[0.74]	-0.117	[0.21]	0.349	[0.19]	0.213	[0.70]	-0.214	[0.20]	0.268	[0.18]	1.257	[0.70]	-0.142	[0.19]	0.574	[0.18]
Strangle N 5%	0.060	[0.40]	-0.118	[0.00]	2.473	[0.00]	0.059	[0.41]	-0.119	[0.00]	2.619	[0.00]	0.532	[0.42]	-0.107	[0.00]	2.650	[0.00]
Strangle F 10%	1.495	[0.44]	-0.029	[0.61]	0.162	[0.28]	1.543	[0.43]	-0.034	[0.58]	0.208	[0.25]	1.745	[0.44]	-0.040	[0.61]	0.337	[0.27]
Strangle N 10%	-0.114	[0.58]	-0.084	[0.00]	2.431	[0.00]	-0.191	[0.59]	-0.088	[0.00]	2.620	[0.00]	-0.293	[0.60]	-0.095	[0.00]	3.014	[0.00]
FN straddle ATM	1.579	[0.38]	0.157	[0.31]	0.287	[0.23]	1.602	[0.38]	0.212	[0.31]	0.412	[0.22]	1.858	[0.37]	0.181	[0.33]	0.517	[0.22]
FN strangle 5%	1.342	[0.57]	0.258	[0.11]	0.491	[0.15]	1.485	[0.57]	0.208	[0.13]	0.427	[0.15]	1.714	[0.53]	0.213	[0.14]	0.555	[0.15]
FN strangle 10%	1.381	[0.51]	0.294	[0.01]	0.936	[0.07]	1.372	[0.53]	0.253	[0.01]	0.708	[0.07]	1.569	[0.49]	0.196	[0.02]	0.615	[0.08]
CS Call ATM	1.674	[0.31]	-0.005	[0.95]	0.136	[0.30]	1.765	[0.32]	-0.011	[0.95]	0.180	[0.27]	2.005	[0.31]	-0.011	[0.95]	0.288	[0.28]
CS Put ATM	1.646	[0.34]	0.017	[0.86]	0.141	[0.30]	1.647	[0.33]	0.050	[0.88]	0.221	[0.28]	1.889	[0.33]	0.051	[0.86]	0.354	[0.29]
CS Call 5%	1.431	[0.60]	0.066	[0.34]	0.262	[0.24]	1.536	[0.59]	0.059	[0.34]	0.290	[0.22]	1.796	[0.60]	0.043	[0.35]	0.346	[0.24]
CS Put 5%	1.697	[0.28]	0.019	[0.83]	0.139	[0.31]	1.749	[0.29]	-0.037	[0.83]	0.188	[0.28]	2.002	[0.28]	-0.010	[0.81]	0.282	[0.28]
CS Call 10%	0.820	[0.84]	0.148	[0.03]	0.958	[0.08]	1.003	[0.87]	0.136	[0.04]	0.969	[0.08]	1.192	[0.90]	0.104	[0.04]	0.863	[0.07]
CS Put 10%	1.795	[0.22]	0.062	[0.33]	0.248	[0.23]	1.782	[0.22]	0.006	[0.34]	0.174	[0.22]	2.012	[0.21]	-0.006	[0.35]	0.282	[0.22]

Table 8: Optimal Investment in Option Strategies with Higher-Frequency Crashes

In this table we study the portfolio allocation of a power-utility investor with CRRA of 2 to each of the option strategies when the frequency of market crashes is higher than what was observed in the sample. To do that, we augment the data set with an extra one or two data points corresponding to the return of each strategy on October of 1987. This corresponds to a doubling or tripling of the frequency of such crashes in the sample. We repeat the CRRA utility maximization of Table 6. For each scenario, the table reports the weight on each of the option strategies, and the increase in certainty equivalent for the investor relative to investing 100% of the wealth in the market alone. Bootstrapped p-values, under the null hypothesis that $w_{mkt}^0 = 1$, $w_{str}^0 = 0$, and $\Delta CE^0 = 0$, are shown in brackets. Options and futures closing prices were sampled monthly between January 1985 and May 2001. The data is provided by the Chicago Mercantile Exchange through the Institute for Financial Markets. All options are American.

	1 crash						2 crashes						3 crashes					
	w_{mkt}		w_{str}		ΔCE $\times 100$		w_{mkt}		w_{str}		ΔCE $\times 100$		w_{mkt}		w_{str}		ΔCE $\times 100$	
Call F ATM	0.754	[0.87]	0.100	[0.25]	0.369	[0.18]	0.175	[0.49]	0.124	[0.14]	0.399	[0.18]	-0.260	[0.29]	0.143	[0.08]	0.497	[0.12]
Call N ATM	1.185	[0.75]	0.038	[0.30]	0.313	[0.23]	0.703	[0.79]	0.046	[0.23]	0.287	[0.22]	0.340	[0.49]	0.052	[0.18]	0.335	[0.17]
Call F 5%	1.459	[0.62]	0.023	[0.79]	0.153	[0.28]	0.868	[0.96]	0.044	[0.54]	0.092	[0.32]	0.434	[0.59]	0.060	[0.42]	0.123	[0.28]
Call N 5%	2.068	[0.12]	-0.027	[0.11]	0.437	[0.16]	1.598	[0.29]	-0.024	[0.13]	0.251	[0.20]	1.254	[0.59]	-0.021	[0.17]	0.178	[0.21]
Call F 10%	1.249	[0.71]	0.040	[0.43]	0.275	[0.23]	0.754	[0.86]	0.051	[0.27]	0.238	[0.25]	0.378	[0.54]	0.059	[0.25]	0.278	[0.21]
Call N 10%	2.122	[0.10]	-0.030	[0.04]	0.626	[0.12]	1.657	[0.28]	-0.027	[0.04]	0.423	[0.14]	1.316	[0.59]	-0.024	[0.05]	0.336	[0.14]
Put F ATM	-1.755	[0.07]	-0.247	[0.01]	0.927	[0.05]	-1.666	[0.08]	-0.210	[0.03]	0.582	[0.10]	-1.583	[0.10]	-0.183	[0.07]	0.411	[0.12]
Put N ATM	-1.715	[0.06]	-0.167	[0.00]	1.688	[0.01]	-2.056	[0.02]	-0.164	[0.00]	1.438	[0.01]	-2.291	[0.01]	-0.160	[0.00]	1.309	[0.01]
Put F 5%	-0.236	[0.30]	-0.105	[0.11]	0.478	[0.14]	0.056	[0.41]	-0.067	[0.26]	0.158	[0.23]	0.263	[0.53]	-0.039	[0.52]	0.044	[0.30]
Put N 5%	0.181	[0.41]	-0.046	[0.01]	1.012	[0.06]	-0.208	[0.20]	-0.045	[0.02]	0.744	[0.07]	-0.466	[0.12]	-0.043	[0.02]	0.603	[0.06]
Put F 10%	1.714	[1.00]	0.001	[0.87]	0.135	[0.27]	1.924	[0.58]	0.023	[0.64]	0.110	[0.31]	2.077	[0.39]	0.039	[0.30]	0.235	[0.20]
Put N 10%	0.310	[0.32]	-0.026	[0.06]	0.616	[0.11]	0.377	[0.39]	-0.017	[0.20]	0.221	[0.21]	0.013	[0.51]	-0.015	[0.42]	0.047	[0.29]
Straddle F ATM	0.936	[0.96]	-0.168	[0.15]	0.415	[0.17]	0.686	[0.76]	-0.135	[0.24]	0.197	[0.24]	0.508	[0.50]	-0.107	[0.34]	0.107	[0.28]
Straddle N ATM	0.691	[0.85]	-0.174	[0.01]	1.137	[0.05]	0.370	[0.43]	-0.156	[0.01]	0.831	[0.05]	0.140	[0.24]	-0.143	[0.02]	0.675	[0.06]
Strangle F 5%	1.104	[0.74]	-0.117	[0.21]	0.349	[0.19]	0.911	[0.98]	-0.072	[0.39]	0.105	[0.28]	0.769	[0.77]	-0.040	[0.61]	0.025	[0.33]
Strangle N 5%	0.060	[0.40]	-0.118	[0.00]	2.473	[0.00]	-0.268	[0.16]	-0.113	[0.00]	2.045	[0.00]	-0.492	[0.05]	-0.108	[0.00]	1.787	[0.00]
Strangle F 10%	1.495	[0.44]	-0.029	[0.61]	0.162	[0.28]	1.294	[0.69]	0.003	[0.93]	0.025	[0.35]	1.146	[0.84]	0.026	[0.75]	0.021	[0.34]
Strangle N 10%	-0.114	[0.58]	-0.084	[0.00]	2.431	[0.00]	-0.333	[0.24]	-0.075	[0.00]	1.758	[0.00]	-0.490	[0.08]	-0.068	[0.00]	1.384	[0.00]
FN straddle ATM	1.579	[0.38]	0.157	[0.31]	0.287	[0.23]	1.178	[0.76]	0.157	[0.29]	0.172	[0.26]	0.878	[0.89]	0.156	[0.29]	0.143	[0.24]
FN strangle 5%	1.342	[0.57]	0.258	[0.11]	0.491	[0.15]	0.973	[0.97]	0.268	[0.09]	0.393	[0.16]	0.691	[0.72]	0.272	[0.09]	0.370	[0.15]
FN strangle 10%	1.381	[0.51]	0.294	[0.01]	0.936	[0.07]	0.938	[1.00]	0.269	[0.04]	0.698	[0.08]	0.623	[0.63]	0.252	[0.03]	0.595	[0.08]
CS Call ATM	1.674	[0.31]	-0.005	[0.95]	0.136	[0.30]	1.268	[0.66]	-0.007	[0.92]	0.028	[0.34]	0.964	[0.98]	-0.009	[0.92]	0.005	[0.34]
CS Put ATM	1.646	[0.34]	0.017	[0.86]	0.141	[0.30]	1.237	[0.69]	0.020	[0.78]	0.033	[0.34]	0.932	[0.96]	0.022	[0.84]	0.009	[0.33]
CS Call 5%	1.431	[0.60]	0.066	[0.34]	0.262	[0.24]	1.022	[1.00]	0.072	[0.23]	0.189	[0.28]	0.717	[0.69]	0.077	[0.21]	0.192	[0.21]
CS Put 5%	1.697	[0.28]	0.019	[0.83]	0.139	[0.31]	1.279	[0.62]	0.003	[0.98]	0.025	[0.35]	0.962	[0.97]	-0.011	[0.91]	0.002	[0.36]
CS Call 10%	0.820	[0.84]	0.148	[0.03]	0.958	[0.08]	0.427	[0.50]	0.157	[0.01]	0.974	[0.06]	0.127	[1.00]	0.163	[0.01]	1.045	[0.03]
CS Put 10%	1.795	[0.22]	0.062	[0.33]	0.248	[0.23]	1.314	[0.58]	0.030	[0.63]	0.053	[0.31]	0.982	[1.00]	0.009	[0.88]	0.003	[0.34]

Table 9: Transaction Costs

This table shows the mean and median bid-ask spread as a percentage of the mid-point price for the index options and the daily mean and median volume, in number of contracts. The results are presented for options with maturities closest to 45, 120, and 180 days, and moneynesses ATM, 5%, and 10% OTM. The data comes from the OptionMetrics database, covering European options on the S&P500 index traded at the Chicago Board Options Exchange in the period between January, 1996 and December, 2002.

	10%	PUT 5%	ATM	ATM	CALL 5%	10%		10%	PUT 5%	ATM	ATM	CALL 5%	10%
<i>mean percentage spread</i>							<i>median percentage spread</i>						
N=45	0.148	0.092	0.059	0.055	0.118	0.315	0.135	0.087	0.056	0.051	0.100	0.229	
120	0.081	0.057	0.038	0.033	0.058	0.094	0.078	0.055	0.037	0.030	0.054	0.087	
F=180	0.063	0.045	0.032	0.026	0.043	0.068	0.061	0.043	0.032	0.024	0.038	0.063	
<i>mean volume</i>							<i>median volume</i>						
N=45	654.8	858.6	993.8	795.4	675.2	455.6	123	247	244	134	165	77	
120	350.1	372.6	405.9	235.3	233.3	273.3	20	62	64	5	6	9	
F=180	229.2	244.9	282.1	252.1	157.7	163.6	5	18	19	1	1	0	

Table 10: Impact of Transaction Costs on Option Strategies' Returns

In this table we analyze the impact of the bid-ask spread on the strategies' returns. For each strategy, we report the mean return, the standard deviation of the return and the Sharpe ratio assuming that the trades are done at midpoint prices (left part of the table) or using the bid price when options are sold and the ask price when options are bought (right part of the table). The data comes from the OptionMetrics database, covering European options on the S&P500 index traded at the Chicago Board Options Exchange in the period between January, 1996 and December, 2002.

	<i>mid-price</i>			<i>bid-to-ask</i>		
	mean	std	<i>SR</i>	mean	std	<i>SR</i>
Put F ATM	0.070	0.423	0.166	0.030	0.432	0.070
Put N ATM	0.184	1.071	0.172	0.140	1.114	0.125
Put F 5%	0.062	0.472	0.131	0.009	0.489	0.018
Put N 5%	0.426	1.211	0.352	0.374	1.292	0.289
Put F 10%	0.120	0.461	0.261	0.052	0.485	0.108
Put N 10%	0.571	1.028	0.555	0.502	1.183	0.425
	mean	std	<i>SR</i>	mean	std	<i>SR</i>
Straddle F ATM	0.018	0.143	0.126	-0.020	0.147	-0.137
Straddle N ATM	0.152	0.286	0.533	0.103	0.293	0.351
Strangle F 5%	0.020	0.199	0.102	-0.032	0.207	-0.153
Strangle N 5%	0.281	0.522	0.538	0.210	0.555	0.379
Strangle F 10%	0.025	0.288	0.087	-0.046	0.308	-0.149
Strangle N 10%	0.557	0.452	1.232	0.479	0.494	0.970

Table 11: Margin “Haircut”

For each option, this table reports the mean, median, standard deviation, minimum, and maximum of the margin haircut, which is the ratio of the margin to the option’s value at the beginning of the month when the trade is implemented, $(M_t - V_0)/V_0$. The CBOE margins (Panel A) are used as a proxy for the margin requirement for customers, and the CME margins (Panel B) as a proxy for the proprietary accounts of a clearing house member. Options and futures closing prices were sampled daily between January 1985 and May 2001. The data is provided by the Chicago Mercantile Exchange through the Institute for Financial Markets. All options are American.

Panel A: CBOE Customers

	mean	median	std	min	max
Put F ATM	4.5	3.7	2.5	0.9	16.5
Put N ATM	7.3	6.4	3.7	2.2	40.9
Put F 5%	5.5	4.3	3.3	1.2	23.0
Put N 5%	21.4	10.9	32.8	2.0	441.4
Put F 10%	8.8	6.3	8.5	1.3	65.1
Put N 10%	48.7	20.8	86.3	2.3	875.0
Straddle F ATM	2.6	2.3	1.4	0.6	8.5
Straddle N ATM	3.9	3.5	1.8	1.1	12.2
Strangle F 5%	3.5	2.8	2.3	0.8	16.0
Strangle N 5%	12.7	7.4	19.5	1.1	243.4
Strangle F 10%	5.0	3.7	4.4	0.8	35.4
Strangle N 10%	24.4	12.6	42.5	1.5	559.3

Panel B: CME Member

	mean	median	std	min	max
Put F ATM	1.6	1.1	1.3	0.3	9.8
Put N ATM	2.6	2.0	2.2	0.3	11.5
Put F 5%	2.0	1.4	1.7	0.3	11.0
Put N 5%	8.7	3.7	15.7	0.3	154.9
Put F 10%	2.9	1.8	3.4	0.4	35.5
Put N 10%	16.4	4.9	35.8	0.2	515.9
Straddle F ATM	0.8	0.6	0.7	0.1	4.1
Straddle N ATM	1.3	0.9	1.1	0.1	5.9
Strangle F 5%	1.2	0.8	0.9	0.2	5.7
Strangle N 5%	4.9	2.4	9.1	0.1	77.6
Strangle F 10%	1.7	1.1	1.6	0.2	15.0
Strangle N 10%	9.3	3.4	20.5	0.1	205.2

Table 12: Impact of Margin Calls on Option Strategies' Returns

This table analyzes the impact of margin calls on the returns to a portfolio of the risk-free asset and the option strategies. We assume that the strategy is implemented at the beginning of each month. If it is not possible to open a position for the desired amount (w_{st} as percentage of wealth) because minimum margin requirements are in excess of wealth, the option strategy weight is recalculated so that the margin due by the investor does not exceed 90% of the available wealth. The amount invested in the risk-free asset is posted as margin (possibly in excess of the minimum required), and assumed to return the risk-free rate. During the month, if the investor faces a margin call due to an adverse movement in the S&P 500, it is met up to availability in the cash account. If and when the investment in the risk-free rate account is exhausted by margin calls, the option position is liquidated, and a loss is realized. We proceed by opening a new position wherein the option strategy weight is recalculated so that the margin due by the investor does not exceed 90% of the available wealth at the end of the day. In the table we report the mean return, the Sharpe ratio of the portfolio return with and without taking into account margin calls. For each strategy we also report the number of times it was necessary to recompute the initial option weight at the inception of the strategy (out), the number of times the account was closed within the month (cl), and the average effective option weight (w_{st}^*). In the last two columns we also report the mean return and Sharpe ratio taking into account both margin calls and transaction costs. As a measure of transaction cost we use the the average return difference reported in Table 10. The CBOE margins (Panel A) are used as a proxy for the margin requirement for customers, and the CME margins (Panel B) as a proxy for the proprietary accounts of a clearing house member. Options and futures closing prices were sampled daily between January 1985 and May 2001. The data is provided by the Chicago Mercantile Exchange through the Institute for Financial Markets. All options are American.

Panel A: CBOE Customer

	no margin			margin					margin + TC	
	mean	SR	w_{st}	mean	SR	out	cl	w_{st}^*	mean	SR
Put F ATM	0.013	0.199	-11.1%	0.010	0.115	12	16	-10.9%	0.007	0.075
Put N ATM	0.026	0.290	-9.2%	0.020	0.196	31	10	-8.7%	0.016	0.153
Put F 5%	0.012	0.211	-8.5%	0.009	0.121	11	18	-8.4%	0.006	0.074
Put N 5%	0.022	0.336	-4.6%	0.013	0.194	66	33	-3.8%	0.010	0.151
Put F 10%	0.004	0.113	-2.8%	0.001	0.014	4	1	-2.8%	-0.001	-0.015
Put N 10%	0.014	0.304	-2.6%	0.005	0.056	82	10	-1.9%	0.002	0.029
Straddle F ATM	0.012	0.183	-23.3%	0.009	0.109	19	11	-22.6%	-0.001	-0.007
Straddle N ATM	0.025	0.255	-19.0%	0.019	0.189	46	17	-17.5%	0.010	0.097
Strangle F 5%	0.010	0.161	-18.2%	0.006	0.086	20	46	-17.4%	-0.004	-0.050
Strangle N 5%	0.045	0.454	-11.7%	0.017	0.239	100	123	-7.9%	0.010	0.127
Strangle F 10%	0.003	0.089	-7.8%	0.003	0.081	8	9	-7.7%	-0.002	-0.064
Strangle N 10%	0.041	0.530	-8.0%	0.010	0.093	121	56	-4.7%	0.005	0.049

Panel B: CME Member

	no margin			margin					margin + TC	
	mean	SR	w_{st}	mean	SR	out	cl	w_{st}^*	mean	SR
Put F ATM	0.013	0.199	-11.1%	0.013	0.196	1	2	-11.1%	0.010	0.146
Put N ATM	0.026	0.290	-9.2%	0.024	0.230	4	3	-9.1%	0.020	0.188
Put F 5%	0.012	0.211	-8.5%	0.011	0.173	0	2	-8.5%	0.008	0.121
Put N 5%	0.022	0.336	-4.6%	0.018	0.313	28	11	-4.2%	0.016	0.263
Put F 10%	0.004	0.113	-2.8%	0.001	0.015	0	1	-2.8%	-0.001	-0.014
Put N 10%	0.014	0.304	-2.6%	0.008	0.095	39	18	-2.3%	0.005	0.060
Straddle F ATM	0.012	0.183	-23.3%	0.012	0.180	0	1	-23.3%	0.003	0.038
Straddle N ATM	0.025	0.255	-19.0%	0.024	0.245	1	4	-18.9%	0.015	0.150
Strangle F 5%	0.010	0.161	-18.2%	0.010	0.162	1	1	-18.2%	0.001	0.012
Strangle N 5%	0.045	0.454	-11.7%	0.028	0.277	43	17	-10.0%	0.022	0.206
Strangle F 10%	0.003	0.089	-7.8%	0.003	0.087	1	0	-7.8%	-0.002	-0.057
Strangle N 10%	0.041	0.530	-8.0%	0.022	0.192	58	23	-6.4%	0.017	0.145

Table 13: Impact of Margin Calls on Option Strategies' Portfolio Weights

This table analyzes the impact of margin calls on the portfolio weight of an investor who faces an investment opportunity set composed by the risk-free asset and the option strategies. We numerically maximize the investor's utility and solve for the portfolio weights when the strategies' returns are constrained by the margin calls. We assume that the strategy is implemented at the beginning of each month. If it is not possible to open a position for the desired amount (w_{st} as percentage of wealth) because minimum margin requirements are in excess of wealth, the option strategy weight is recalculated so that the margin due by the investor does not exceed 90% of the available wealth. The amount invested in the risk-free asset is posted as margin (possibly in excess of the minimum required), and assumed to return the risk-free rate. During the month, if the investor faces a margin call due to an adverse movement in the S&P 500, it is met up to availability in the cash account. If and when the investment in the risk-free rate account is exhausted by margin calls, the option position is liquidated, and a loss is realized. We proceed by opening a new position wherein the option strategy weight is recalculated so that the margin due by the investor does not exceed 90% of the available wealth at the end of the day. Options and futures closing prices were sampled daily between January 1985 and May 2001. The data is provided by the Chicago Mercantile Exchange through the Institute for Financial Markets. All options are American.

	no margin		CBOE margin		CME margin	
	w_{st}	CE ×100	w_{st}	CE ×100	w_{st}	CE ×100
Put F ATM	-11.1%	0.767	-10.2%	0.714	-11.1%	0.745
Put N ATM	-9.2%	1.545	-7.3%	1.389	-8.4%	1.501
Put F 5%	-8.5%	0.753	-6.5%	0.683	-7.8%	0.744
Put N 5%	-4.6%	1.433	-4.0%	1.135	-4.7%	1.310
Put F 10%	-2.8%	0.214	-2.4%	0.202	-2.7%	0.213
Put N 10%	-2.6%	0.962	-1.7%	0.458	-1.9%	0.582
Straddle F ATM	-23.3%	0.666	-19.3%	0.596	-20.6%	0.654
Straddle N ATM	-19.0%	1.382	-16.4%	1.255	-17.9%	1.368
Strangle F 5%	-18.2%	0.534	-16.4%	0.491	-17.9%	0.526
Strangle N 5%	-11.7%	2.933	-10.8%	1.076	-10.0%	2.096
Strangle F 10%	-7.8%	0.174	-7.6%	0.194	-7.3%	0.169
Strangle N 10%	-8.0%	2.892	-3.3%	0.847	-5.3%	1.606